

Tutorial: Build Assembly `as1` from a Kit of Parts

The Assembly Structure tutorial (`../as1-oc-214/README.md`) explored an assembly that already exists. This one goes the other direction: starting from the five loose, un-positioned parts produced by `extract_prototypes.py` (see that script, or ask about it if you don't have the file), we rebuild the real `as1-oc-214` assembly from scratch -- exercising sibling-assembly creation, shared instances, and nearly every method the Position dialog offers, all in one continuous build.

What you'll need

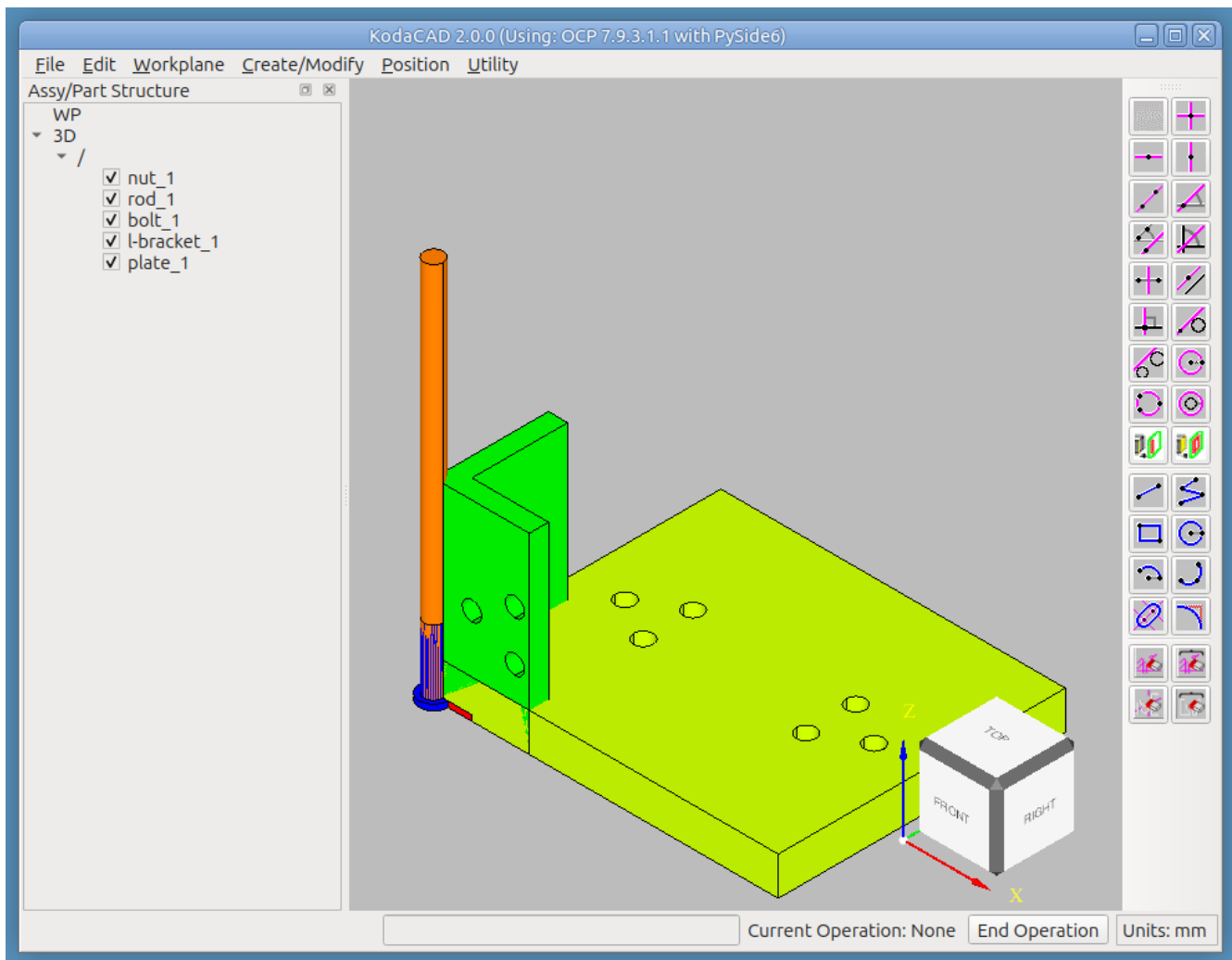
`as1-oc-214-kit-of-parts.stp` -- five parts (`plate` , `l-bracket` , `bolt` , `nut` , `rod`), each sitting at the origin where it was originally modeled, all superimposed on top of each other.

What this exercises

Import STEP into an empty session; creating a new assembly directly at the root; drag-reparenting parts into an assembly; the Position dialog's full method set (Dynamic, Mate Align with all three constraints -- Mate, Align, and Align Axis -- and 2 Points, including Ctrl+Shift circle-center picking); shared-instance creation for both a part and an assembly, including verifying an instance moved independently of its sibling; workplane-on-face combined with Align-Gizmo-to-Active-Workplane for a rotation about a specific axis; and the calculator's Dist tool used mid-build to correct a positioning error rather than just to inspect a finished model.

Step 1 -- Import the file into an empty session

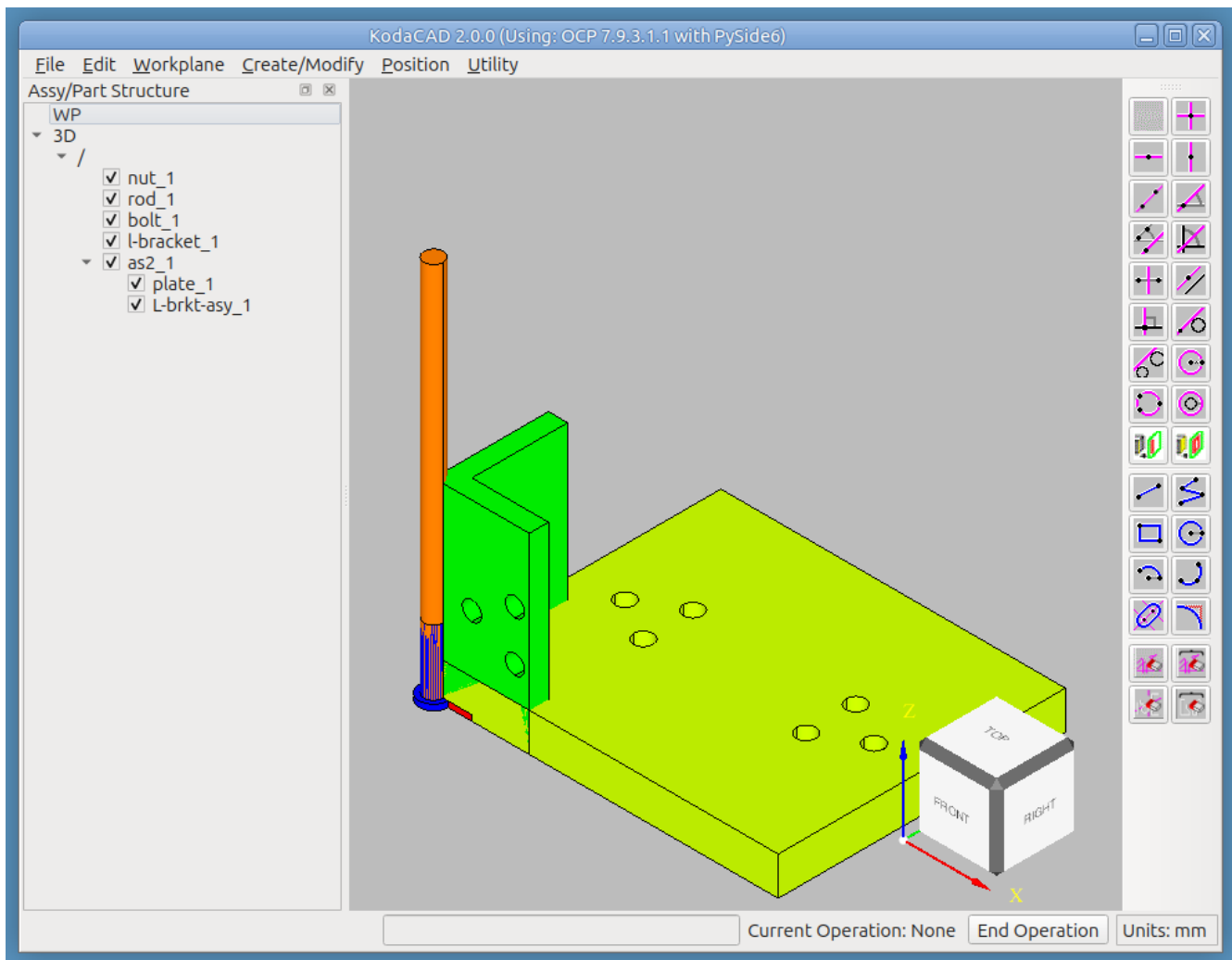
File -> Import STEP, choose `as1-oc-214-kit-of-parts.stp` .



Exercises: Import STEP into a genuinely empty session -- the parts land exactly where they were modeled, all overlapping at the origin, which is the whole reason a "kit of parts" file is useful for this kind of tutorial.

Step 2 -- Begin creating the assembly tree

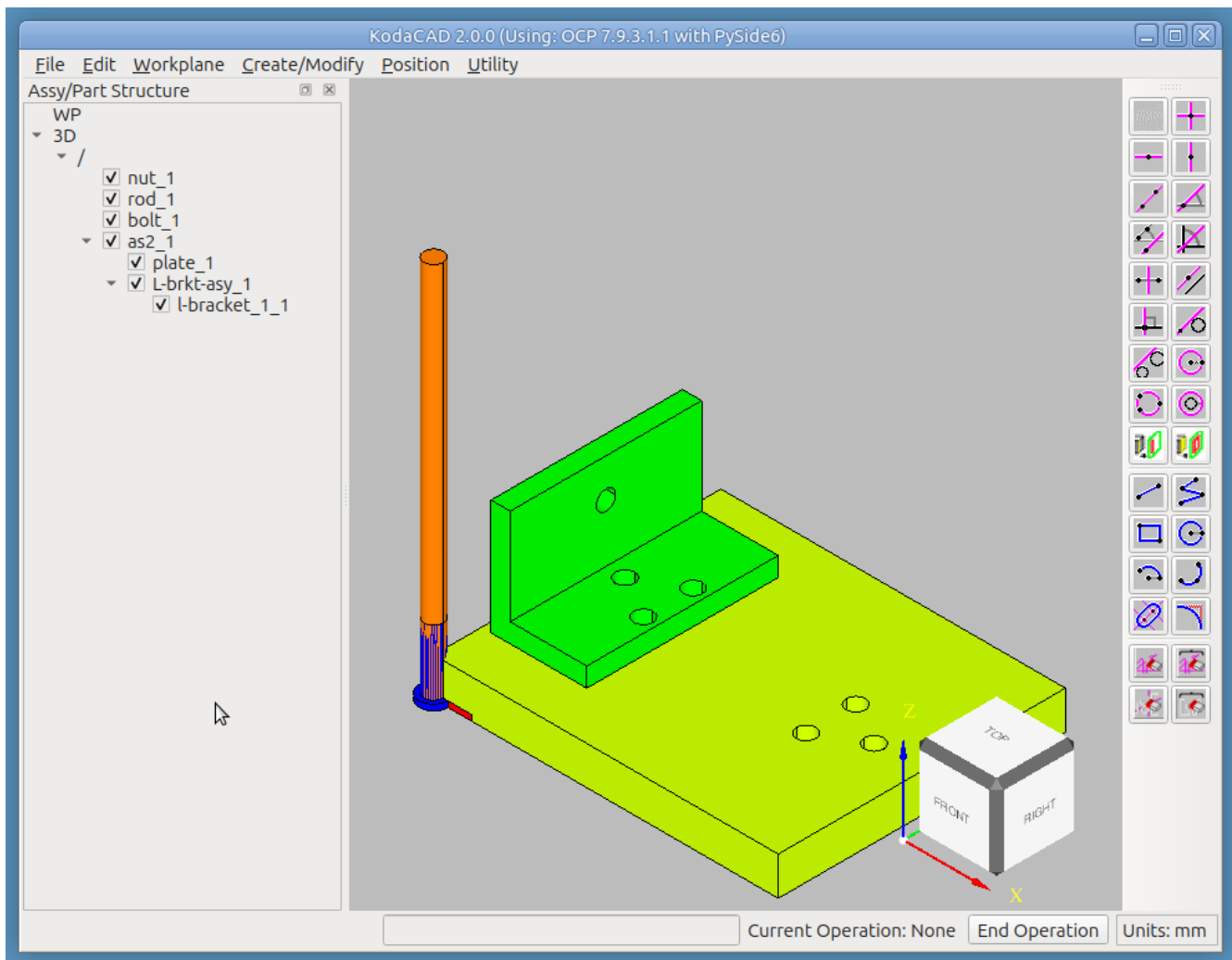
- RMB click / and select **Create New Assembly**. Name it `as2` (to differentiate it from the original assembly, named `as1`).
- Drag `plate` into the new `as2` assembly in the tree. It's already in the correct position, so no further move is needed.
- RMB click `as2` and select **Create New Assembly**. Name it `L-brkt-asy`. (An upper-case `L` here makes it more obvious the bracket's shape resembles the letter it's named after.)



Exercises: creating a new assembly directly at the root in a fresh session, and creating a second, nested assembly under the first -- confirming both are siblings under their own parent, not accidentally nested into each other.

Step 3 -- Add and position the L-bracket

- Drag l-bracket into L-brkt-asy in the tree.
- Select l-bracket in the tree, then **Position -> Position Selected**.
 - Choose method **Mate Align**.
 1. Constraint **Mate** -- click the bottom face of the bracket, then the top face of the plate.
 2. Constraint **Align Axis** -- click the cylindrical face of one of the holes in the bracket, then the cylindrical face of the corresponding hole in the plate.
 3. Constraint **Align** (a third constraint, not always strictly needed) -- click the bracket's back (vertical) face, then the plate's left side, to guarantee they're parallel.

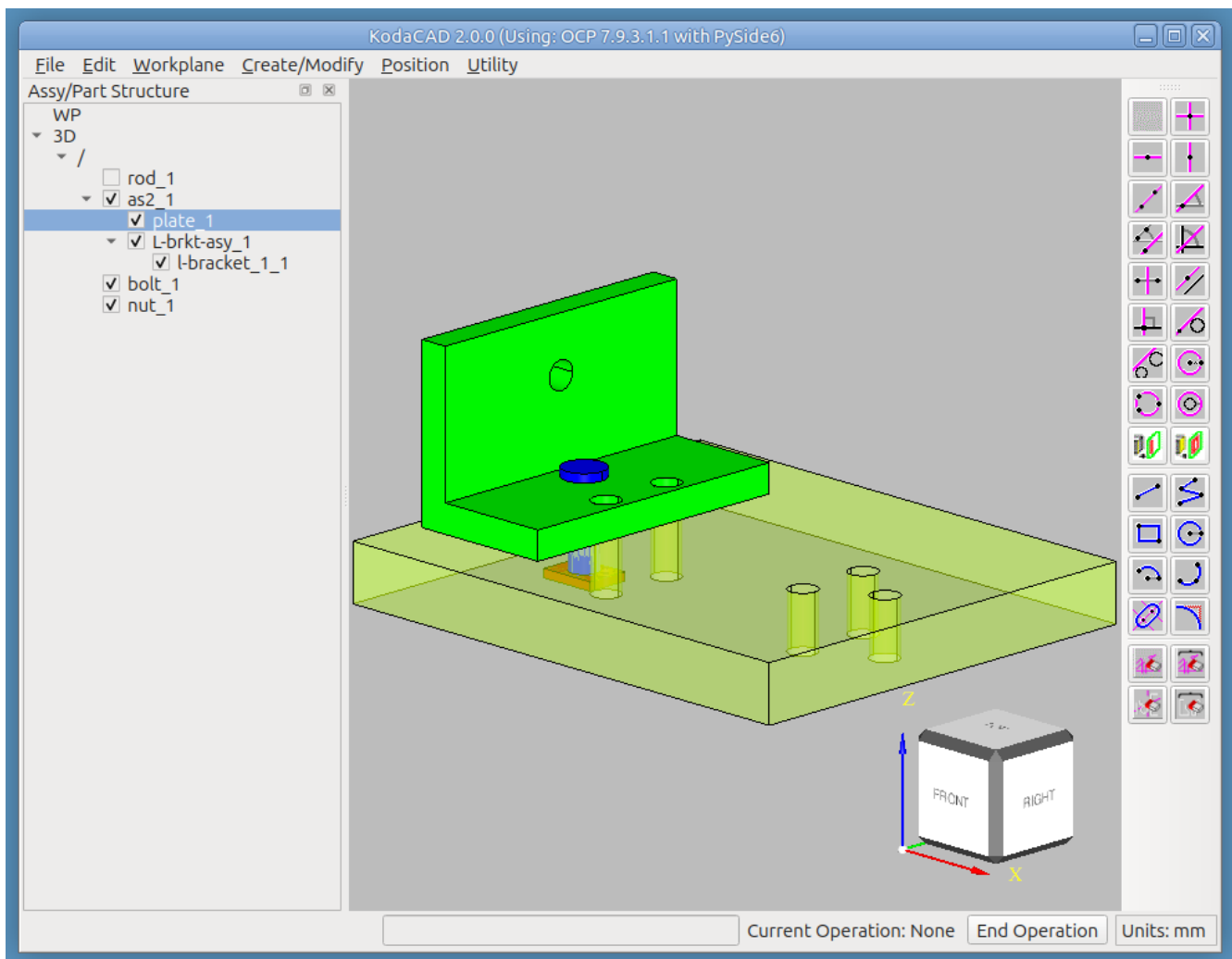


Exercises: Mate Align with all three constraint types in one sequence -- Mate, then Align Axis, then Align. Align Axis is valid here as the second constraint applied (it can be first or second, but never the last of the three).

Step 4 -- Position the bolt and nut

- Hide the rod first, to make the bolt easier to pick.
- Select `bolt` in the tree, then **Position -> Position Selected**.
 - Choose method **Mate Align**.
 1. Constraint **Align Axis** -- click the bolt's cylindrical face, then the cylindrical surface of the target hole in the L-bracket.
 2. Constraint **Mate** -- click the underside of the bolt head, then the top face of the plate.
 3. Click Done.
- Position the nut in two stages -- first out of the way, then precisely:
 - Select `nut` in the tree, then **Position -> Position Selected**.

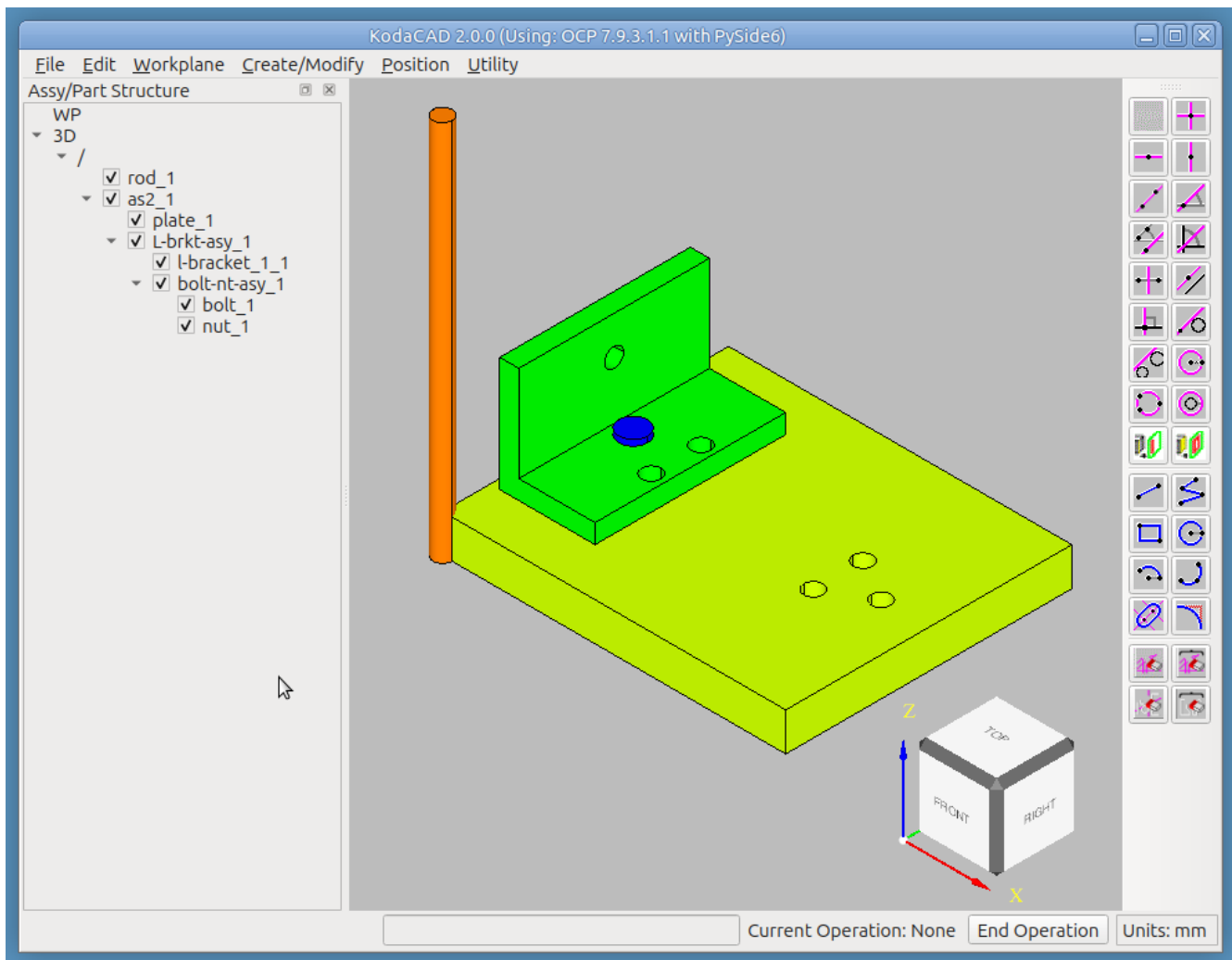
- Stage 1, method **Dynamic** -- drag the nut a short distance away from the plate using the gizmo's handles, just far enough to make picking easier.
- Stage 2, method **2 Points**.
 1. First point: hold **Ctrl+Shift** and click the top edge of the nut's cylindrical hole. (Ctrl+Shift while picking a circular edge finds its center, not a point on the edge itself.)
 2. Second point: hold **Ctrl+Shift** and click the edge of the matching hole in the underside of the plate.
 3. Click Done.



Exercises: Mate Align with Align Axis used as the FIRST constraint this time (valid either way, per Step 3's own note); Dynamic used deliberately as a rough, intermediate move rather than a final one; 2 Points with Ctrl+Shift circle-center picking on a genuinely 3D edge.

Step 5 -- Create the bolt-nut assembly

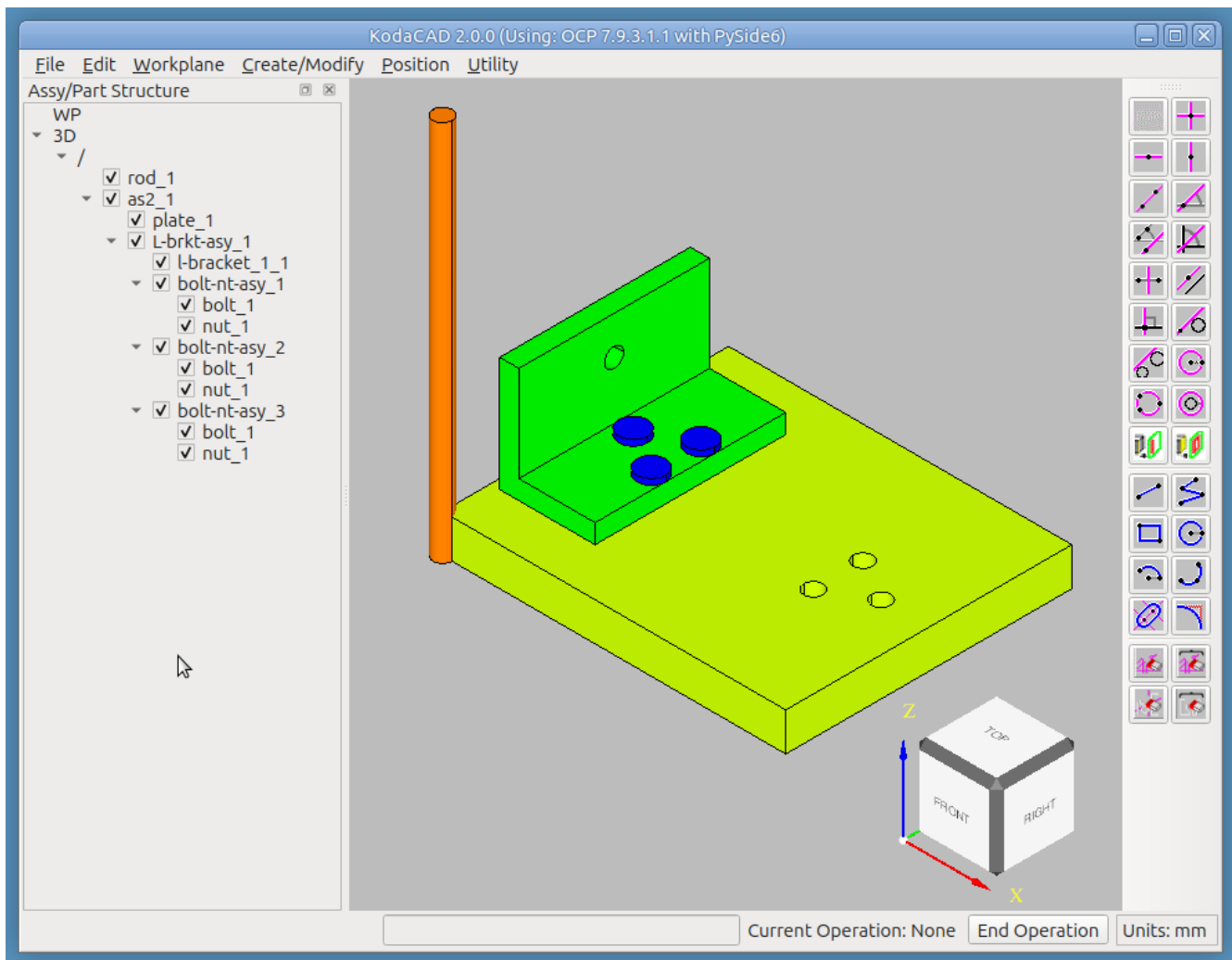
- RMB click L-brkt-asy and select **Create New Assembly**. Name it bolt-nut-asy .
- Drag bolt into the new bolt-nut-asy .
- Drag nut into bolt-nut-asy .



Exercises: grouping two already-positioned, unrelated parts into a new sub-assembly after the fact, rather than creating the assembly first and populating it before positioning anything.

Step 6 -- Create two more instances of the bolt-nut assembly

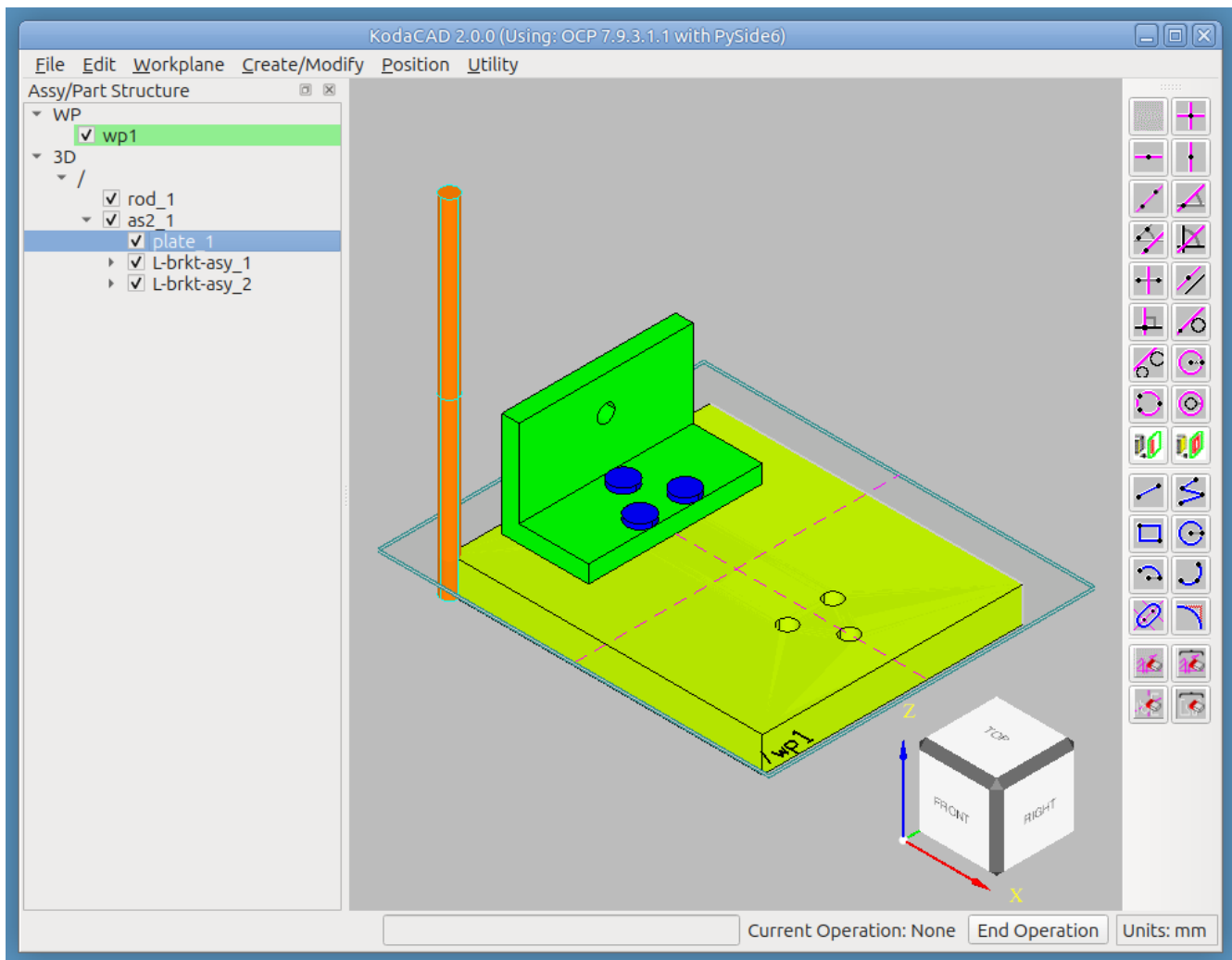
- RMB click `bolt-nut-asy_1` and select **Create Shared Instance**. Name the new one `bolt-nut-asy_2`. It starts out exactly superimposed on the original – the only way to tell it's there at all is to toggle the two instances' checkboxes independently in the tree.
- Move `bolt-nut-asy_2` to the next hole:
 - With both instances hidden, select the new one.
 - **Position -> Position Selected**, method **2 Points**.
 - Holding Ctrl+Shift, click the circular edge of the hole the assembly currently occupies, then the target hole.
- Repeat to create `bolt-nut-asy_3` and position it in the third hole.



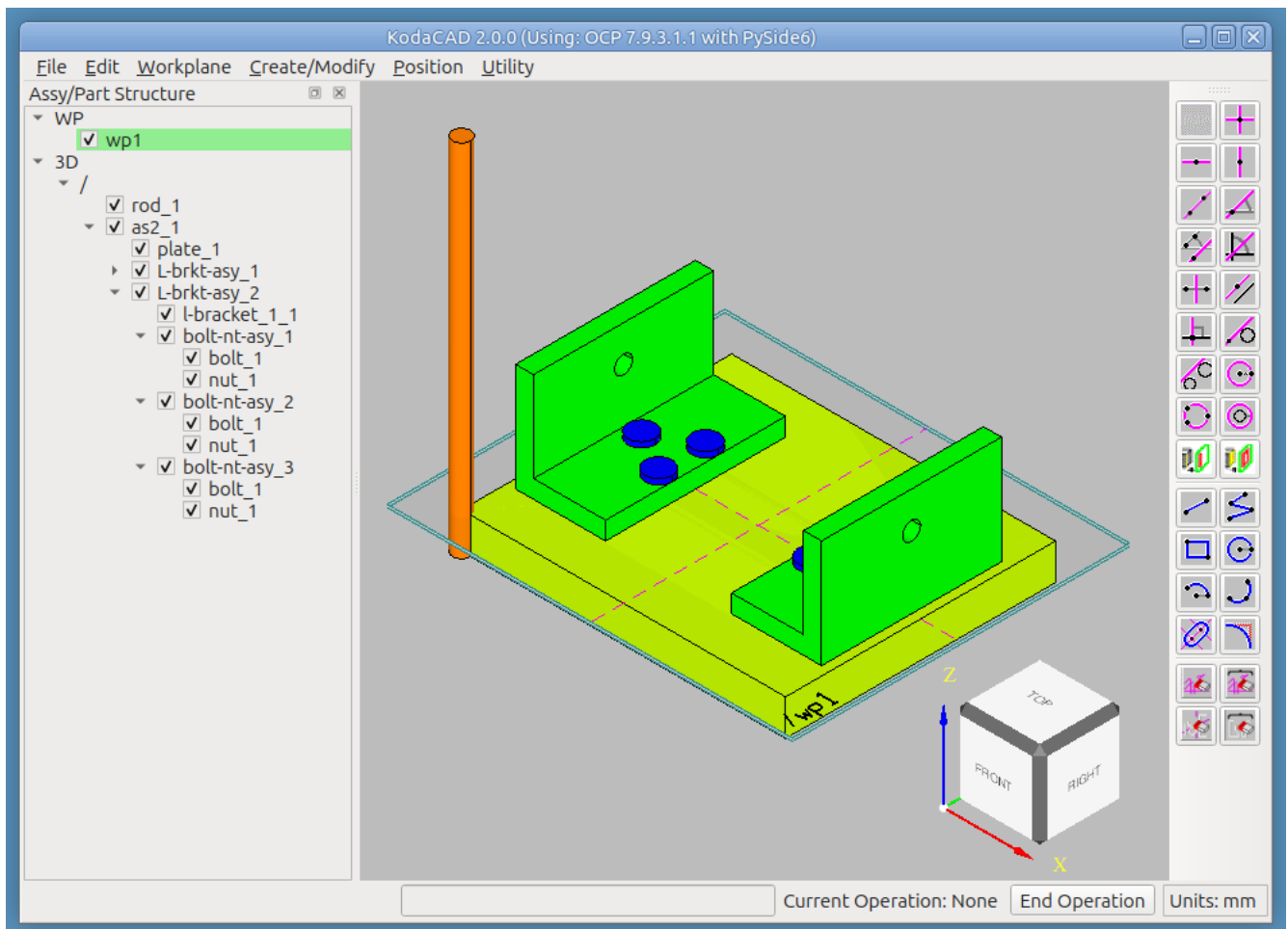
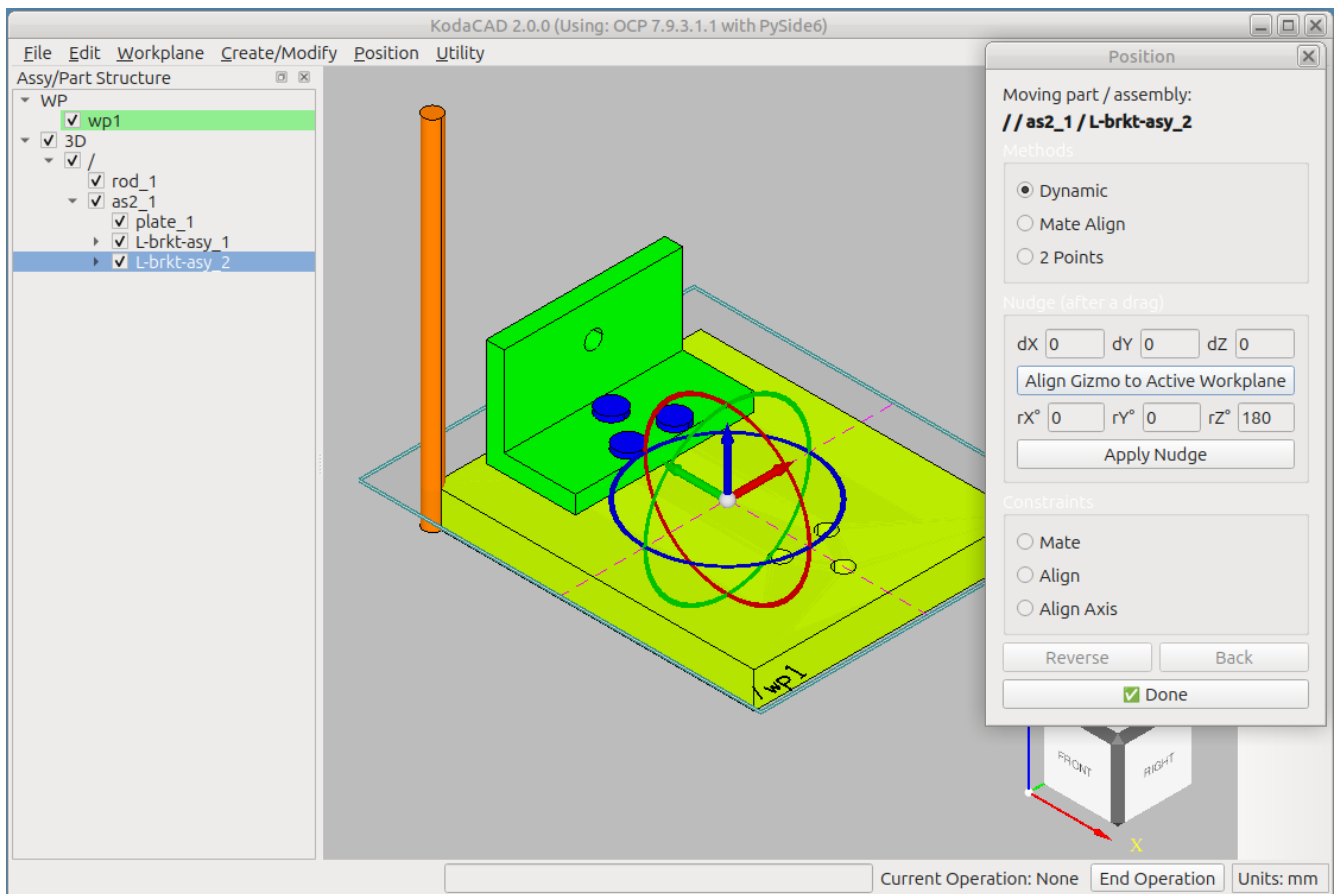
Exercises: shared-instance creation for an assembly (not just a leaf part), and 2 Points used to reposition a whole shared assembly instance rather than a single component.

Step 7 -- Create and position the second L-bracket assembly

- RMB click L-brkt-asy_1 and select **Create Shared Instance**. Name it L-brkt-asy_2 .
- This one needs a 180-degree rotation about a Z axis through the center of the plate -- so first, build a workplane there: **Workplane -> On Face**, click the plate's top face, then one of its side faces. (Picking the first face places the workplane's own origin at that face's geometric center. The second face determines the workplane's +U direction.)



- Select L-brkt-asy_2 in the tree, then **Position -> Position Selected**.
 - Choose method **Dynamic**.
 - Click **Align Gizmo to Active Workplane**.
 - Enter 180 in the **rZ** field, then **Apply Nudge** to preview the move.
 - If it looks wrong, **Back** and try again; if it looks right, **Done**.

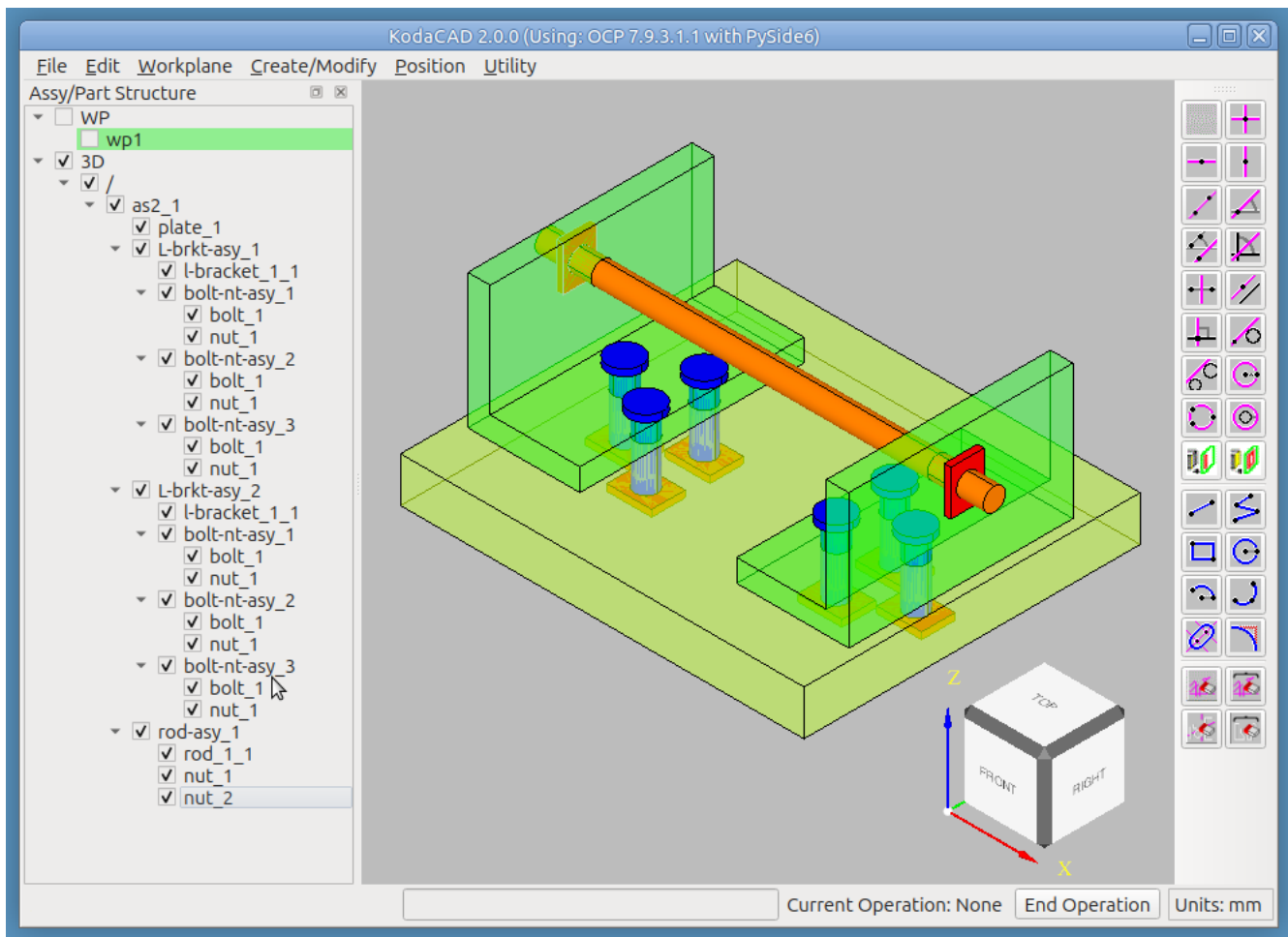


Exercises: workplane-on-face combined with Align Gizmo to Active Workplane, so a nudge rotation happens about a specific, meaningful axis (through the plate's center) rather than the part's own default pivot.

Step 8 -- Create the rod assembly

The rod assembly needs the rod moved into its correct position and dropped into a new assembly -- in either order.

- RMB click `as2` and select **Create New Assembly**. Name it `rod-asy`. Drag `rod` into it.
- Select `rod`, then **Position -> Position Selected**.
 - Method **Mate Align**.
 - Constraint **Align Axis** -- click the rod's cylindrical face, then the cylindrical face of the target hole in one of the L-brackets.
 - Constraint **Align** -- click an end face of the rod, then the corresponding end face of the plate.
 - **Done**.
- At this point the rod isn't quite centered -- it extends farther past one L-bracket than the other. **Utility -> Calculator**, use **Dist** to measure exactly how far the rod needs to move in X to be centered on the assembly.
- Re-enter the Position dialog and nudge the rod in X by that amount.
- Create a shared instance of one of the nuts:
 1. Reposition it onto the rod, using whichever method you prefer.
 2. Drag it into `rod-asy` in the tree.
- Create a second shared instance of the nut and reposition it to the opposite end of the rod.



Exercises: Align Axis + Align (two constraints, not the full three of Mate Align) as a deliberately different combination than earlier steps; using the calculator's Dist tool mid-build to correct a real positioning error, rather than only for post-build inspection; two more shared instances of a single-part prototype (the nut), each positioned independently.

Notes for whoever runs this next

- If any menu label, dialog control name, or button text has drifted from what's written here, that's worth fixing in this document directly.
- See the Assembly Structure tutorial ([../as1-oc-214/README.md](#)) for loading and exploring the *original* as1-oc-214.stp -- a natural point of comparison once this one is finished, to confirm the rebuilt assembly matches the real thing structurally, not just visually.
- See the Lathe tutorial ([../lathe/README.md](#)) for organizing and navigating a large, messily-imported assembly -- renaming, sibling assemblies, and reparenting an assembly that already has a parent to a different one.